MICHAEL SENDKER

Build & Test Engineer

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WHO I AM

Platform, build and test engineer: I find the pain points in how a company builds and tests its software and engineer them away, especially in old code. For five years I've done that for AiDiffract (formerly NanoDiffract), a decades-old, 2M+ line C++ metrology package in a Windows shop, and I'm the person its teams come to for pipelines, build machines, test infrastructure, and the problems that don't belong to anyone else. Self-taught, with a background in philosophy and classics; I also led the company's developer book club.

EXPERIENCE

Onto Innovation, Metrology Business Unit

Build & Test Engineer (Contract) · 2021 – Current

`Python` — `C++` — `C#` — `C` — `pytest` — `PostgreSQL` — `Qt` — `Azure DevOps` — `RHEL`

- Designed and built, **over five years**, the regression-testing system **every AiDiffract engineer (~30) relies on**: a pytest-based tool that generates per-build reference results, stores them in PostgreSQL, and runs PR validation and post-merge regressions on the cluster. Moved its database and recipe packages to RHEL.
- Built its desktop GUI (PySide6, Cython), in use at **five leading chipmakers** and rolling out to every Onto customer.
- Built tooling on top of it: automated bisection across official builds that finds the build behind a result change **in minutes** (once a manual, build-by-build hunt), a benchmark harness with statistical significance testing, and a memory monitor: a small HTTP server in C on each blade that the head node polls for memory use.
- Automated recovery of wedged and lost nodes in the cluster management system for application developers, and upgraded its agent to .NET 8. Traced erratic behavior on 80 nodes to a faulty port on the server rack.
- Own the build infrastructure: provision and maintain the build machines, write the teams' Azure DevOps pipelines, and led compiler and dependency upgrades (including Intel MKL) and a build-system redesign. Cleared ~**7,000 compiler warnings** across AiDiffract by hand.
- Rewrote AiDiffract's neural net inference as a **Python C extension** so ML engineers could analyze RCWA results.
- Helped build the AutoML product's test suite, adding speed-ups, automated PDF reports, and faster file reading.
- Now: a leaner results schema, node memory data in a redesigned history app, and new test recipe packaging.

PROJECTS

bignews.day

Personal Project · 2024 – Current

`Python` — `C` — `SQLAlchemy` — `Ollama` — `Whisper` — `scikit-learn`

- Follows how the news is told across 145 outlets: front pages, wire copy, TV chyrons, radio and podcasts (transcribed locally), claims on social media, and fact-checks, clustered into stories and multi-part sagas.
- Classifies the recurring narratives in political claims, in an index built bottom-up and grounded in folklore scholarship; local models propose classifications and a model trained on my own decisions scores them.
- Every model runs locally on one 10 GB GPU, chosen by testing candidates against my own judgments, not public benchmarks; a methods log dates every change in measurement.
- Built the 2024 version (Maudlin News) by hand, including multithreaded scrapers and **my own story clustering algorithm** (after trying HDBSCAN and k-means): TF-IDF cosine similarity thresholded into a graph whose connected components are stories, searched by a C extension for a 10x speedup. Restarted in October 2026 with AI assistance.

EARLIER EXPERIENCE

HtmMbs LLC, Software Developer: perpetual inventory and allocation tools for an Odoo ERP. 2021

Healthy Brands, LLC, Contract Developer: an Amazon BuyBox monitor and ask.fiteyes.com. 2021

Euler Scientific, Contract: the GUI for a dermatology laser system (C#, WPF). 2020 – 2021

Glendale Painting, Office Manager & Technology Director: website, marketing data tools. 2017 – 2020

EDUCATION & LANGUAGES

Florida State University, Philosophy and Classics: Greek and Latin; Philosophy Club president. 2007 – 2010

Languages: English (native), French (fluent), Spanish (rudimentary), Latin (for fun).
Free time: Daily runner and cyclist; reading epic poetry, learning Latin, and studying compilers.